

Test Report

Number: SHAH01530119

Applicant: CHIZHOU HAIZHIBAO CHILDREN PRODUCTS
CO., LTD
MATANG INDUSTRIAL PARK, DINGQIAO TOWN,
QINGYANG COUNTY, CHIZHOU CITY
Attn: RITA CAO

Date: 17 Feb, 2023

Sample Description:

One (1) group of submitted sample said to be :

Item Name : Toy Car.
Item No. : Audi Q5/HZB-108.
Labelled Age Group : 3-8 years.
Packaging Provided By Applicant : Yes.
Goods Exported To : EU.
Country Of Origin : CHINA

Tests Conducted:

As requested by the applicant, for details refer to attached page(s).

Prepared And Checked By:
For Intertek Testing Services Wuxi Ltd.



Peter Chen
General Manager



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Conclusion:

<u>Tested Samples</u>	<u>Standard</u>	<u>Result</u>
Submitted Sample set	EN71-1: 2014+ A1: 2018 for Mechanical And Physical Properties	Pass
Submitted Sample set	EN71-2: 2020 Flammability Test	Pass
Tested component(s) of submitted sample(s)	EN 71-3:2019+A1:2021 on migration of certain elements	Pass
Tested component(s) of submitted sample(s)	Cadmium content requirement in Commission Regulation (EU) No. 494/2011 of 20 May 2011, (EU) No. 835/2012 of 18 September 2012 and (EU) No. 2016/217 of 16 February 2016 Amending Annex XVII Items 23 of the Reach Regulation (EC) No. 1907/2006	Pass
Tested component(s) of submitted sample(s)	Phthalates content requirement in Annex XVII Item 51 of the REACH Regulation (EC) No. 1907/2006 & Amendment No. 552/2009 & Amendment Commission Regulation (EU) 2018/2005 (formerly known as Directive 2005/84/EC)	Pass
Tested component(s) of submitted sample(s)	Phthalates content requirement in Annex XVII Items 52 of the REACH Regulation (EC) No. 1907/2006 & Amendment No. 552/2009 (formerly known as Directive 2005/84/EC)	Pass
Submitted Sample set	EN IEC 62115:2020+A11:2020- Safety of Electric Toys Excluding the Clause 15.4, 19, Annex D ,Annex E, Annex I , Annex J	Pass (Subjected to Remark enclosed)
Submitted Sample set	Australian / New Zealand Standard AS/NZS ISO 8124.1:2019+Amd.1:2020+Amd.2:2020 Safety Aspects Related to Mechanical And Physical Properties	Pass
Submitted Sample set	Australian / New Zealand Standard on Safety of Toys AS/NZS 8124.2: 2016 Flammability Test	Pass
Tested component(s) of submitted sample(s)	Australian Competition and Consumer Act 2010 with Consumer Protection Notice No. 11, 2011 - Permanent ban on children's products with Diethylhexyl Phthalate (DEHP)	Pass
Tested component(s) of submitted sample(s)	Australian / New Zealand standard on safety of toys AS/NZS ISO 8124 part 3 : 2021 for toxic elements test	Pass
Tested component(s) of submitted sample(s)	Azocolourants content requirement in Annex XVII Item 43 of the REACH Regulation (EC) No. 1907/2006 & Amendment (EC) No. 552/2009 and (EU) 2020/2096	Pass

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General Manager



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Tests Conducted

1 Mechanical and Physical Test

As Per European Standard on Safety of Toys EN71-1: 2014+ A1: 2018.

Applicant's Specified Age Group for Testing: For 3 - 8 Years.

The submitted samples were undergone the following abuse tests:		
Test	Clause	Parameter
Torque test	8.3	0.34 Nm
Tension test	8.4.2.1	90 N
Protective components	8.4.2.3	60 N
Drop test	8.5	850 mm x 5times
Tip over test	8.6	Three times
Impact test	8.7	1 kg
Compression test	8.8	110 N
Flexibility of metallic wires	8.13	70 N

Clause	Testing items	Assessment
4	General requirements	
4.1	Material	P
4.2	Assembly	P
4.3	Flexible plastic sheeting	NA
4.4	Toy bags	NA
4.5	Glass	NA
4.6	Expanding materials	NA
4.7	Edges	P
4.8	Points and metallic wires	P
4.9	Protruding parts	P
4.10	Parts moving against each other	P
4.11	Mouth actuated toys and other toys intended to be put in the mouth	NA
4.12	Balloons	NA
4.13	Cords of toy kites and other flying toys	NA
4.14	Enclosures	NA
4.15	Toys intended to bear the mass of a child	P
4.16	Heavy immobile toys	NA
4.17	Projectile toys	NA
4.18	Aquatic toys and inflatable toys	NA
4.19	Percussion caps specifically designed for use in toys and toys using percussion caps	NA
4.20	Acoustics	P
4.21	Toys containing a non-electrical heat source	NA
4.22	Small balls	NA
4.23	Magnets	NA
4.24	Yo-yo balls	NA
4.25	Toys attached to food	NA



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Clause	Testing items	Assessment
4.26	Toy disguise costumes	NA
4.27	Flying toys	NA
5	Toys intended for children under 36 months	
5.1	General requirements	NA
5.2	Soft-filled toys and soft-filled parts of a toy	NA
5.3	Plastic sheeting	NA
5.4	Cords, chains and electrical cables in toys	NA
5.5	Liquid filled toys	NA
5.6	Speed limitation of electrically-driven ride-on toys	NA
5.7	Glass and porcelain	NA
5.8	Shape and size of certain toys	NA
5.9	Toys comprising monofilament fibres	NA
5.10	Small balls	NA
5.11	Play figures	NA
5.12	Hemispheric-shaped toys	NA
5.13	Suction cups	NA
5.14	Straps intended to be worn fully or partially around the neck	NA
5.15	Sledges with cords for pulling	NA
6	Packaging	P
7	Warnings, markings and instructions for use	
7.1	General	P
7.2	Toys not intended for children under 36 months	P
7.3	Latex balloons	NA
7.4	Aquatic toys	NA
7.5	Functional toys	NA
7.6	Hazardous sharp functional edges and points	NA
7.7	Projectile toys	NA
7.8	Imitation protective masks and helmets	NA
7.9	Toy kites	NA
7.10	Roller skates, inline skates and skateboards and certain other ride-on toys	P
7.11	Toys intended to be strung across a cradle, cot, or perambulator	NA
7.12	Liquid-filled teethingers	NA
7.13	Percussion caps specifically designed for use in toys	NA
7.14	Acoustics	NA
7.15	Toy bicycles	NA
7.16	Toys intended to bear the mass of a child	NA
7.17	Toys comprising monofilament fibres	NA
7.18	Toy scooters	NA
7.19	Rocking horses and similar toys	NA
7.20	Magnetic/electrical experimental sets	NA
7.21	Toys with electrical cables exceeding 300 mm in length	NA
7.22	Toys with cords or chains intended for children of 18 months and over but under 36 months	NA
7.23	Toys intended to be attached to a cradle, cot or perambulator	NA
7.24	Sledges with cords for pulling	NA



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Clause	Testing items	Assessment
7.25	Flying toys	NA
7.26	Improvised projectiles	NA

Remark: P = Pass

NA = Not Applicable

Remark: Additional information according to the Toy Safety Directives 2009/48/EC requirement. These information also appears as a note within the EN 71 but are not standard requirements:

1. Marking

The manufacturer's and importer's name, registered trade name or registered trade mark, the address and the CE-marking shall be indicated on the toy or, where that is not possible, on its packaging or in a document accompany the toy. In addition, manufacturers shall ensure that their toys bear a type, batch, serial or model number or other element allowing their identification, or where the size or nature of the toy does not allow it, that the required information is provided on the packaging or in a document accompanying the toy.

After checking, it was found that:

	Toy	Packaging
Manufacturer's name	Present	Present
Manufacturer's address	Present	Present
Importer's name	Absent	Absent
Importer's address	Absent	Absent
Product identification code	Present	Present
CE-marking	Absent	Present

Below is additional information checking according to the UK Toy (Safety) Regulations requirement.

Marking

The manufacturer's and importer's name, registered trade name or registered trademark, the address and type, batch, serial or model number or other element allowing their identification shall be indicated on the product itself.

After checking, it was found that:

	Toy	Packaging
Name of authorised representative in Great Britain	Absent	Absent
Address of authorised representative in Great Britain	Absent	Absent
Product identification code	Present	Present

With reference to the guidance of using UKCA marking from 1 January 2021 by the Department for Business, Energy and Industrial Strategy published on 1 September 2020.

After checking UKCA marking, it was found that:

	Toy	Packaging
UKCA marking	Absent	Absent

Date Sample Received : 16 Dec, 2022 & Jan 10, 2023

Testing Period : 16 Dec, 2022 to 23 Jan 10, 2023



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Tests Conducted

2 **Flammability Test**

As per European Standard on Safety of Toys EN71-2: 2020

Clause	Testing Items	Assessment
4.1	General	P
4.2	Toys to be worn on the head	
4.2.2	Beards, moustaches, wigs, etc., made from pile or flowing elements which protrude 50 mm or more from the surface of the toy	NA
4.2.3	Beards, moustaches, wigs, etc., made from pile or flowing elements which protrude less than 50 mm from the surface of the toy	NA
4.2.4	Full or partial moulded head masks	NA
4.2.5	Toys to be worn on the head	NA
4.3	Toy Disguise Costumes and Toys Intended to be Worn by a Child in Play	NA
4.4	Toys Intended to be Entered by a Child	NA
4.5	Soft Filled Toys	NA

Remark : P = Pass

NA = Not Applicable

Date Sample Received : 16 Dec, 2022

Testing Period : 16 Dec, 2022 to 23 Dec, 2022



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Tests Conducted

3 19 Toxic Element Migration Test
(A) Test Result

As per EN 71-3:2019+A1:2021 and followed by Inductively Coupled Plasma Atomic Emission Spectrometry, Inductively Coupled Argon Mass Spectrometry, Ion Chromatography- Inductively Coupled Plasma-Mass Spectrometry, Ion Chromatography with UV-VIS and Gas Chromatographic - Mass Spectrometry.

Category (III): Scraped-off toy material

Element	Result (mg/kg)						Reporting Limit (mg/kg)	Limit (mg/kg)
	(1)	(2)	(3)	(4)	(5)	(6)		
Aluminium (Al)	ND	ND	ND	ND	ND	ND	300	28130
Antimony (Sb)	ND	ND	ND	ND	ND	ND	10	560
Arsenic (As)	ND	ND	ND	ND	ND	ND	10	47
Barium (Ba)	ND	ND	ND	ND	ND	ND	10	18750
Boron (B)	ND	ND	ND	ND	ND	ND	50	15000
Cadmium (Cd)	ND	ND	ND	ND	ND	ND	5	17
Chromium (III) (Cr III) ⁺⁺	ND	ND	ND	ND	ND	ND	10	460
Chromium (VI) (Cr VI) ⁺⁺	ND	ND	ND	ND	ND	ND	0.025	0.053
Cobalt (Co)	ND	ND	ND	ND	ND	ND	10	130
Copper (Cu)	ND	ND	ND	ND	ND	ND	10	7700
Lead (Pb)	ND	ND	ND	ND	ND	ND	10	23
Manganese (Mn)	ND	ND	ND	ND	ND	ND	10	15000
Mercury (Hg)	ND	ND	ND	ND	ND	ND	10	94
Nickel (Ni)	ND	ND	ND	ND	ND	ND	10	930
Selenium (Se)	ND	ND	ND	ND	ND	ND	10	460
Strontium (Sr)	ND	ND	ND	ND	ND	ND	100	56000
Tin (Sn)	ND	ND	ND	ND	ND	ND	2.5	180000
Organic tin ⁺⁺	ND	ND	ND	ND	ND	ND	5	12
Zinc (Zn)	ND	ND	ND	ND	ND	ND	100	46000



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Element	Result (mg/kg)						Reporting Limit (mg/kg)	Limit (mg/kg)
	(7)	(8)	(9)	(10)	(11)	(12)		
Aluminium (Al)	ND	ND	ND	ND	ND	ND	300	28130
Antimony (Sb)	ND	ND	ND	ND	ND	ND	10	560
Arsenic (As)	ND	ND	ND	ND	ND	ND	10	47
Barium (Ba)	ND	ND	ND	ND	ND	ND	10	18750
Boron (B)	ND	ND	ND	ND	ND	ND	50	15000
Cadmium (Cd)	ND	ND	ND	ND	ND	ND	5	17
Chromium (III) (Cr III) **	ND	ND	ND	ND	ND	ND	10	460
Chromium (VI) (Cr VI) **	ND	ND	ND	ND	ND	ND	0.025	0.053
Cobalt (Co)	ND	ND	ND	ND	ND	ND	10	130
Copper (Cu)	ND	ND	ND	ND	ND	ND	10	7700
Lead (Pb)	ND	ND	ND	ND	ND	ND	10	23
Manganese (Mn)	ND	ND	ND	ND	ND	ND	10	15000
Mercury (Hg)	ND	ND	ND	ND	ND	ND	10	94
Nickel (Ni)	ND	ND	ND	ND	ND	ND	10	930
Selenium (Se)	ND	ND	ND	ND	ND	ND	10	460
Strontium (Sr)	ND	ND	ND	ND	ND	ND	100	56000
Tin (Sn)	ND	ND	ND	ND	ND	ND	2.5	180000
Organic tin **	ND	ND	ND	ND	ND	ND	5	12
Zinc (Zn)	ND	ND	ND	ND	ND	ND	100	46000

Element	Result (mg/kg)						Reporting Limit (mg/kg)	Limit (mg/kg)
	(13)	(14)	(15)	(16)	(17)	(18)		
Aluminium (Al)	ND	ND	ND	ND	ND	ND	300	28130
Antimony (Sb)	ND	ND	ND	ND	ND	ND	10	560
Arsenic (As)	ND	ND	ND	ND	ND	ND	10	47
Barium (Ba)	ND	ND	ND	ND	ND	ND	10	18750
Boron (B)	ND	ND	ND	ND	ND	ND	50	15000
Cadmium (Cd)	ND	ND	ND	ND	ND	ND	5	17
Chromium (III) (Cr III) **	ND	ND	ND	ND	ND	ND	10	460
Chromium (VI) (Cr VI) **	ND	ND	ND	ND	ND#	ND	0.025	0.053
Cobalt (Co)	ND	ND	ND	ND	ND	ND	10	130
Copper (Cu)	ND	ND	ND	ND	ND	ND	10	7700
Lead (Pb)	ND	ND	ND	ND	ND	ND	10	23
Manganese (Mn)	ND	ND	ND	ND	ND	ND	10	15000
Mercury (Hg)	ND	ND	ND	ND	ND	ND	10	94
Nickel (Ni)	ND	ND	ND	ND	ND	ND	10	930
Selenium (Se)	ND	ND	ND	ND	ND	ND	10	460
Strontium (Sr)	ND	ND	ND	ND	ND	ND	100	56000
Tin (Sn)	ND	ND	ND	ND	ND	ND	2.5	180000
Organic tin **	ND	ND	ND	ND	ND	ND	5	12
Zinc (Zn)	ND	ND	ND	ND	ND	ND	100	46000



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Element	Result (mg/kg)						Reporting Limit (mg/kg)	Limit (mg/kg)
	(19)	(20)	(21)	(22)	(23)	(24)		
Aluminium (Al)	ND	ND	ND	ND	ND	ND	300	28130
Antimony (Sb)	ND	ND	ND	ND	ND	ND	10	560
Arsenic (As)	ND	ND	ND	ND	ND	ND	10	47
Barium (Ba)	ND	ND	ND	ND	ND	ND	10	18750
Boron (B)	ND	ND	ND	ND	ND	ND	50	15000
Cadmium (Cd)	ND	ND	ND	ND	ND	ND	5	17
Chromium (III) (Cr III) **	ND	ND	ND	ND	ND	ND	10	460
Chromium (VI) (Cr VI) **	ND	ND	ND	ND	ND	ND	0.025	0.053
Cobalt (Co)	ND	ND	ND	ND	ND	ND	10	130
Copper (Cu)	ND	ND	ND	ND	ND	ND	10	7700
Lead (Pb)	ND	ND	ND	ND	ND	ND	10	23
Manganese (Mn)	ND	ND	ND	ND	ND	ND	10	15000
Mercury (Hg)	ND	ND	ND	ND	ND	ND	10	94
Nickel (Ni)	ND	ND	ND	ND	ND	ND	10	930
Selenium (Se)	ND	ND	ND	ND	ND	ND	10	460
Strontium (Sr)	ND	ND	ND	ND	ND	ND	100	56000
Tin (Sn)	ND	ND	ND	ND	ND	ND	2.5	180000
Organic tin **	ND	ND	ND	ND	ND	ND	5	12
Zinc (Zn)	ND	ND	ND	ND	ND	ND	100	46000

Element	Result (mg/kg)			Reporting Limit (mg/kg)	Limit (mg/kg)
	(25)	(26)	(27)		
Aluminium (Al)	ND	ND	ND	300	28130
Antimony (Sb)	ND	ND	ND	10	560
Arsenic (As)	ND	ND	ND	10	47
Barium (Ba)	ND	ND	ND	10	18750
Boron (B)	ND	ND	ND	50	15000
Cadmium (Cd)	ND	ND	ND	5	17
Chromium (III) (Cr III) **	ND	ND	ND	10	460
Chromium (VI) (Cr VI) **	ND	ND#	ND#	0.025	0.053
Cobalt (Co)	ND	ND	ND	10	130
Copper (Cu)	ND	ND	ND	10	7700
Lead (Pb)	ND	ND	ND	10	23
Manganese (Mn)	ND	ND	ND	10	15000
Mercury (Hg)	ND	ND	ND	10	94
Nickel (Ni)	ND	ND	ND	10	930
Selenium (Se)	ND	ND	ND	10	460
Strontium (Sr)	ND	ND	ND	100	56000
Tin (Sn)	ND	ND	ND	2.5	180000
Organic tin **	ND	ND	ND	5	12
Zinc (Zn)	ND	ND	ND	100	46000



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Remark : mg/kg = milligram per kilogram
 ++ = Unless the test results were marked with "#" or "Δ", Chromium (III) & Chromium (VI) and Organic tin contents were not directly determined and were derived from migration results of total chromium and tin respectively.
 - Organic tin test result was expressed as tributyl tin.
 ND = Not detected (less than reporting limit)
 # = Confirmation of Chromium (VI) test was performed on the tested component. And the reported value of migration of Chromium (III) = migration value of total Chromium – migration value of Chromium(VI).
 Δ = Confirmation test was performed on the tested component. The reported value was the sum of the migration values of Dimethyl tin, Methyl tin, Butyl tin, Dibutyl tin, Tributyl tin, Tetra-butyl tin, n-Octyl tin, Di-n-octyl tin, Di-n-propyl tin, Diphenyl tin and Triphenyl tin after converted to Tributyl tin by calculation. Other Organic tin compounds may be also be present in sample as stated in EN 71-3:2019+A1:2021.

Tested component(s): See component list in the last section of this report.

(B) Categories of various toy materials

Category I: Dry, brittle, powder like or pliable

Solid toy material from which powder-like material is released during playing and semi-solid materials that may also leave residues on the hands during play. The material can be ingested. Contamination of the hands with the material may contribute to the oral exposure of the material. (e.g. the cores of colouring pencils, chalk, crayons, modelling clays and plaster).

Category II: Liquid or sticky

Fluid or viscous toy material, which can be ingested or to which dermal exposure may occur during playing. (e.g. liquid paints, finger paints, liquid ink in pens, glue sticks, slimes, bubble solution).

Category III: Scraped-off

Solid toy material with or without a coating, which can be ingested as a result of biting, tooth scraping, sucking or licking. (e.g. coatings, lacquers, plastics, paper, textiles, glass, ceramic, metallic, wooden, bone, leather and other materials).

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Tests Conducted

4 Cadmium (Cd) Content

With reference to methods IEC 62321, acid digestion method was used and total Cadmium content was determined by Inductively Coupled Argon Plasma Spectrometry.

<u>Tested Component</u>	<u>Result in %</u>
(1)	ND
(2)	ND
(3)	ND
(4)	ND
(5)	ND
(6)	ND
(7)	ND
(8)	ND
(9)	ND
(10)	ND
(11)	ND
(12)	ND
(13)	ND
(14)	ND
(15)	ND
(16)	ND
(17)	ND
(18)	ND
(19)	ND
(20)	ND
(21)	ND
(22)	ND
(23)	ND
(24)	ND
(25)	ND
(26)	ND

Requirement:	
<u>Category</u>	<u>Limit (%)</u>
Painted article	0.1
Plastic	0.01
Metal parts of jewellery & hair accessories	0.01

Remark: ND = Not Detected (<0.0005%)

Tested Components: See component list in the last section of this report.

Date Sample Received : 16 Dec, 2022

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Tests Conducted

5 Phthalate Content

With reference to ISO 8124-6: 2018 method A or C, by Gas Chromatographic-Mass Spectrometric (GC-MS) analysis.

Tested Compound	CAS No.	Result (%w/w)							Limit (%w/w)
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(Max.)
Dibutyl phthalate (DBP)	84-74-2	ND	ND	ND	ND	ND	ND	ND	-
Diethyl hexyl phthalate (DEHP)	117-81-7	ND	ND	ND	ND	ND	ND	ND	-
Benzyl butyl phthalate (BBP)	85-68-7	ND	ND	ND	ND	ND	ND	ND	-
Diisobutyl phthalate (DIBP)	84-69-5	ND	ND	ND	ND	ND	ND	ND	-
Sum of DBP,DEHP,BBP and DIBP	--	ND	ND	ND	ND	ND	ND	ND	0.1

Tested Compound	CAS No.	Result (%w/w)							Limit (%w/w)
		(8)	(9)	(10)	(11)	(12)	(13)	(14)	(Max.)
Dibutyl phthalate (DBP)	84-74-2	ND	ND	ND	ND	ND	ND	ND	-
Diethyl hexyl phthalate (DEHP)	117-81-7	ND	ND	ND	ND	ND	ND	ND	-
Benzyl butyl phthalate (BBP)	85-68-7	ND	ND	ND	ND	ND	ND	ND	-
Diisobutyl phthalate (DIBP)	84-69-5	ND	ND	ND	ND	ND	ND	ND	-
Sum of DBP,DEHP,BBP and DIBP	--	ND	ND	ND	ND	ND	ND	ND	0.1

Tested Compound	CAS No.	Result (%w/w)							Limit (%w/w)
		(15)	(16)	(17)	(18)	(19)	(20)	(21)	(Max.)
Dibutyl phthalate (DBP)	84-74-2	ND	ND	ND	ND	ND	ND	ND	-
Diethyl hexyl phthalate (DEHP)	117-81-7	ND	ND	ND	ND	ND	ND	ND	-
Benzyl butyl phthalate (BBP)	85-68-7	ND	ND	ND	ND	ND	ND	ND	-
Diisobutyl phthalate (DIBP)	84-69-5	ND	ND	ND	ND	ND	ND	ND	-
Sum of DBP,DEHP,BBP and DIBP	--	ND	ND	ND	ND	ND	ND	ND	0.1

Tested Compound	CAS No.	Result (%w/w)					Limit (%w/w)
		(22)	(23)	(24)	(25)	(26)	(Max.)
Dibutyl phthalate (DBP)	84-74-2	ND	ND	ND	ND	ND	-
Diethyl hexyl phthalate (DEHP)	117-81-7	ND	ND	ND	ND	ND	-
Benzyl butyl phthalate (BBP)	85-68-7	ND	ND	ND	ND	ND	-
Diisobutyl phthalate (DIBP)	84-69-5	ND	ND	ND	ND	ND	-
Sum of DBP,DEHP,BBP and DIBP	--	ND	ND	ND	ND	ND	0.1

The above limit was quoted according to Annex XVII Item 51 of the REACH Regulation (EC) No. 1907/2006 & Amendment No. 552/2009 & Amendment Commission Regulation (EU) 2018/2005 for phthalate content in articles.

Remark: Detection Limit = 0.01%(w/w)
ND = Not Detected

@ = As requested by the applicant, the surface coatings were tested with the substrate for phthalate test. With the consideration of the dilution factor, the testing result may not represent the result of the individual coatings and substrate.

Tested Components: See component list in the last section of this report.

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Tests Conducted

6 Phthalate Content

With reference to ISO 8124-6: 2018 method A or C, by Gas Chromatographic-Mass Spectrometric (GC-MS) analysis.

Tested Compound	CAS No.	Result (%w/w)							Limit (%w/w)
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(Max.)
Di-n-octyl phthalate (DnOP)	117-84-0	ND	ND	ND	ND	ND	ND	ND	-
Diisononyl phthalate (DINP)	28553-12-0/ 68515-48-0	ND	ND	ND	ND	ND	ND	ND	-
Diisodecyl phthalate (DIDP)	26761-40-0/ 68515-49-1	ND	ND	ND	ND	ND	ND	ND	-
Sum of DINP, DNOP and DIDP	--	ND	ND	ND	ND	ND	ND	ND	0.1

Tested Compound	CAS No.	Result (%w/w)							Limit (%w/w)
		(8)	(9)	(10)	(11)	(12)	(13)	(14)	(Max.)
Di-n-octyl phthalate (DnOP)	117-84-0	ND	ND	ND	ND	ND	ND	ND	-
Diisononyl phthalate (DINP)	28553-12-0/ 68515-48-0	ND	ND	ND	ND	ND	ND	ND	-
Diisodecyl phthalate (DIDP)	26761-40-0/ 68515-49-1	ND	ND	ND	ND	ND	ND	ND	-
Sum of DINP, DNOP and DIDP	--	ND	ND	ND	ND	ND	ND	ND	0.1

Tested Compound	CAS No.	Result (%w/w)							Limit (%w/w)
		(15)	(16)	(17)	(18)	(19)	(20)	(21)	(Max.)
Di-n-octyl phthalate (DnOP)	117-84-0	ND	ND	ND	ND	ND	ND	ND	-
Diisononyl phthalate (DINP)	28553-12-0/ 68515-48-0	ND	ND	ND	ND	ND	ND	ND	-
Diisodecyl phthalate (DIDP)	26761-40-0/ 68515-49-1	ND	ND	ND	ND	ND	ND	ND	-
Sum of DINP, DNOP and DIDP	--	ND	ND	ND	ND	ND	ND	ND	0.1

Tested Compound	CAS No.	Result (%w/w)					Limit (%w/w)
		(22)	(23)	(24)	(25)	(26)	(Max.)
Di-n-octyl phthalate (DnOP)	117-84-0	ND	ND	ND	ND	ND	-
Diisononyl phthalate (DINP)	28553-12-0/ 68515-48-0	ND	ND	ND	ND	ND	-
Diisodecyl phthalate (DIDP)	26761-40-0/ 68515-49-1	ND	ND	ND	ND	ND	-
Sum of DINP, DNOP and DIDP	--	ND	ND	ND	ND	ND	0.1

The above limit was quoted according to Annex XVII Item 52 of the REACH Regulation (EC) No. 1907/2006 & Amendment No. 552/2009 for phthalate content in toys and childcare articles.

Remark: Detection Limit = 0.01%(w/w)
ND = Not Detected



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@ = As requested by the applicant, the surface coatings were tested with the substrate for phthalate test. With the consideration of the dilution factor, the testing result may not represent the result of the individual coatings and substrate.

Tested Components: See component list in the last section of this report.

Date Sample Received : 16 Dec, 2022
Testing Period : 16 Dec, 2022 to 23 Dec, 2022

7 Safety of Electric Toys

As per European Standard on Safety of Electric Toys EN IEC 62115:2020+A11:2020

Applicant's Specified Age Group for Testing: For 3 - 8 Years.

Power source: Remote: 3 V, LR 03 size x 2 pcs,
Vehicle: 12 V, 7 Ah, Lead-acid rechargeable battery x 1 pc

Charger: type: Input 100-240V A.C. Output 12 V D.C. (Provided)
Model: GA09-1201000US

Electric Operated Function: Battery powered LED light, Sound, motion.

Clause	Requirement	Assessment
1	Scope	--
2	Normative reference	--
3	Term and definitions	--
4	General requirement	--
5	General conditions for test	--
6	Criteria for reduced testing	NA
6.1	General	--
6.2	Short-circuit resistance	NA
6.3	Low power electric toys	NA
6.4	Battery circuits	NA
7	Marking and instructions	P
7.1	General	P
7.2	Marking on electric toys	P
7.2.1	Identification	See remark(1)
7.2.2	Electric toys with replaceable batteries	P
7.2.3	Transformer toys and power supply toys	NA
7.2.4	Electric toys with more than one power supply	NA
7.2.5	Electric toys with detachable lamps	NA
7.2.6	Symbols	NA
7.2.7	Durability	P
7.3	Instructions and markings on packaging	P
7.3.1	General	P
7.3.2	Transformer toys and power supply toys	P
7.3.3	Electric toys that are used with replaceable batteries	P
7.3.3.1	General	P
7.3.3.2	Coin batteries	NA
7.3.3.3	Button batteries	NA
7.4	Instructions for electric toys that can be connected to class I equipment	NA
7.5	Instructions for ride-on electric toys	P
7.6	Temperature warnings	NA
8	Power input	NA



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Clause	Requirement	Assessment
9	Heating and abnormal operation	P
9.1	General	P
9.2	Test condition	--
9.3	Normal operation	P
9.4	Normal operation with insulation short-circuited	P
9.5	Abnormal operation with temperature controls made inoperable	NA
9.6	With accessible moving parts locked	P
9.7	Additional transformers and power supplies	NA
9.8	Abnormal supply to electric toys via a USB connection.	NA
9.9	Fault condition in electronic circuits	P
9.10	Compliance criteria	P
10	Electric strength	P
10.1	Electric strength at operating temperature	P
10.2	Electric strength under humid conditions	P
11	Electric toys used in water, electric toys used with liquid and electric toys cleaned with liquid	NA
12	Mechanical strength	P
12.1	Enclosures	P
12.2	Attachment strength	P
13	Construction	P
13.1	Nominal supply voltage	P
13.2	Transformers, power supplies and battery chargers	P
13.3	Thermal cut-outs.	NA
13.4	Batteries	P
13.4.1	Small batteries	P
13.4.2	Other batteries	P
13.4.3	Electrolyte leakage	P
13.4.4	Electric toys placed above a child	NA
13.4.5	Parallel connection of batteries	P
13.4.6	Battery compartment fasteners	P
13.5	Plug and sockets	P
13.6	Charging batteries	P
13.7	Series motors	NA
13.8	Working voltage	NA
13.9	Electric toys connecting to other equipment.	NA
13.10	Speed limitation of ride-on electric toys	P
14	Protection of cords and wires	P
14.1	Edges and moving parts	P
14.2	Fixed parts	NA
15	Components	--
15.1.1	General	P See remark(2)
15.1.2	Switches and automatic controls	NA
15.1.3	Other components	P See remark(2)
15.2	Prohibited components	P
15.3	Transformers and power supplies	NA
15.4	Battery chargers	See remark(3)
15.5	Batteries	NA
16	Screws and connections	P
16.1	Fixings	P
16.2	Connections	P
17	Clearances and creepage distances	P
18	Resistance to heat and fire	P



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Clause	Requirement	Assessment
18.1	Resistance to heat	P
18.2	Resistance to fire	P
18.2.1	General	P
18.2.2	Non-metallic parts	P
18.2.3	Insulating material	P
19	Radiation and similar hazards	See remark(3)
19.1	General	--
19.2	Optical radiation Toys incorporating lasers and or light emitting diodes (LED) or UV emitting lamps shall comply with Annex E. Electric toys incorporating LEDs shall comply with 19.E.2. Electric toys incorporating lasers shall comply with 19.E.3. Electric toys incorporating UV-emitting lamps shall comply with 19.E.4.	See remark(3)
19.3	Other electromagnetic radiation Electric toys with an integrated field source that may produce harmful electromagnetic radiation Measurements methods are given in Annex I.	See remark(3)
Annex A	Experimental sets	NA
Annex B	Needle-flame test	NA
Annex C	Automatic controls and switches	NA
Annex D	Electric toys with protective electronic circuits	See remark(3)
Annex E	Safety of electric toys incorporating optical radiation sources	See remark (3)
Annex F	Flowcharts showing the assessment of optical radiation safety of LEDs in electric toys	--
Annex G	Examples of calculations on LEDs	--
Annex H	Explanation of the principles used for the requirements of Annex E	--
Annex I	Electric toys generating electromagnetic fields (EMF)	See remark (3)
Annex J	Safety of remote controls for electric ride-on toys	See remark (3)
Annex K	Flow charts showing the application of Clause 9	--

Abbreviation: P = Pass

NA = Not Applicable

Remark:

(1)

Only the English version of the marking and instructions were assessed. According to the standard, instruction sheets and other texts required by the standard shall be written in the official language of the country in which the product is to be sold.

Below are additional information according to the requirement in Toy Safety Directive 2009/48/EC relating to marking of toys and do not constitute requirements of this European Standard:

The manufacturer's and importer's name, registered trade name or registered trade mark, the address and type, batch, serial or model number or other element allowing their identification shall be indicated on the toy or, where that is not possible, on its packaging or in a document accompanying the toy.

After checking, it was found that:

	Toy	Packaging
Manufacturer's name	Present	Present
Manufacturer's address	Present	Present
Importer's name	Absent	Absent
Importer's address	Absent	Absent
Product identification code	Present	Present

(2)

Applicant needs to ensure that components used in toys shall comply with the safety requirements specified in the relevant standards.

Applicant needs to ensure that battery charger for toys shall comply with IEC 60335-2-29:2016 and



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Annex AA of that standard.

(3) As requested by the applicant, the Annex D , Annex E, Annex I , Anne J were not assessed.

Date Sample Received : 16 Dec, 2022

Testing Period : 16 Dec, 2022 to 23 Dec, 2022

8 **Physical and Mechanical Tests**

As per the Australian / New Zealand Standard AS/NZS ISO 8124.1:2019+Amd.1:2020+Amd.2:2020 Safety Aspects Related to Mechanical and Physical Properties.

Applicant's Specified Age Group for Testing: For 3 - 8 Years.

The submitted samples were undergone the normal use and the following reasonable foreseeable abuse tests in accordance with the Clause 5.24 of AS/NZS ISO 8124.1:2019+Amd.1:2020+Amd.2:2020 before the assessment of the relevant requirement in Clause 4:

<u>Clause</u>	<u>Test</u>	<u>Parameter</u>
5.24.2	Drop test	4x93±5cm
5.24.3	Tip-over test	3 times
5.24.4	Dynamic strength	--
5.24.5	Torque test	0.45±0.02Nm
5.24.6	Tension test	70±2N
5.24.7	Compression test	136±2N
5.24.8	Flexure test	70±2N

<u>Section</u>	<u>Testing Items</u>	<u>Assessment</u>
4.1	Normal use	P
4.2	Reasonably foreseeable abuse	P
4.3	Material	P
4.4	Small parts	NA
4.5	Shape, size and strength of certain toys	NA
4.6	Edges	P
4.7	Points	P
4.8	Projections	P
4.9	Metal wires and rods	P



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<u>Section</u>	<u>Testing Items</u>	<u>Assessment</u>
4.10	Plastic film or plastic bags in packaging and in toys	P
4.11	Cords	NA
4.12	Folding mechanisms	P
4.13	Holes, clearances and accessibility of mechanisms	P
4.14	Springs	NA
4.15	Stability and overload requirements	P
4.16	Enclosures	NA
4.17	Simulated protective equipment, such as helmets, hats and goggles	NA
4.18	Projectile toys	NA
4.19	Flying toys	NA
4.20	Aquatic toys	NA
4.21	Braking	P
4.22	Toy bicycles	NA
4.23	Speed limitation of electrically driven ride-on toys	P
4.24	Toys containing a heat source	NA
4.25	Liquid-filled toys	NA
4.26	Mouth-actuated toys	NA
4.27	Toy roller skates, toy inline skates and toys skateboards	NA
4.28	Percussion caps specifically designed for use in toys	NA
4.29	Acoustic requirements	P
4.30	Toy scooters	NA
4.31	Magnets and magnetic components	NA
4.32	Yo-yo balls	NA
4.33	Straps intended to be worn fully or partially around the neck	NA
4.34	Sledges and toboggans with cords for pulling	NA



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Tests Conducted

<u>Section</u>	<u>Testing Items</u>	<u>Assessment</u>
4.35	Jaw entrapment in handles and steering wheels	NA
4.36	Assembly	P
Annex B	Safety-labelling guidelines and manufacturer's markings	P
Annex D	Toy gun marking	NA

Remark : P = Pass

NA = Not applicable

Date Sample Received : 16 Dec, 2022 & Jan 10, 2023

Testing Period : 16 Dec, 2022 to Jan 10, 2023

9 **Flammability Test**

As per Australian/New Zealand Standard on Safety of Toys AS/NZS 8124.2: 2016.

<u>Clause</u>	<u>Testing Items</u>	<u>Assessment</u>
4.1	General Requirements	P
4.2	Toys to be Worn on the Head	NA
4.3	Toy Disguise Costumes and Toys Intended to be Worn by a Child in a Play	NA
4.4	Toys Intended to be Entered by a Child	NA
4.5	Soft-Filled Toys	NA

Remark: P = Pass

NA = Not Applicable

Date Sample Received : 16 Dec, 2022

Testing Period : 16 Dec, 2022 to 23 Dec, 2022



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Tests Conducted

10 Diethylhexyl phthalate (DEHP) Content

With reference to ISO 8124-6:2018, and determined by Gas Chromatographic-Mass Spectrometric (GC-MS) analysis.

<u>Tested Component</u>	<u>Result (%w/w)</u>	<u>Limit(%w/w)</u> <u>(MAX.)</u>
(1)	ND	1
(2)	ND	1
(3)	ND	1
(4)	ND	1
(5)	ND	1
(6)	ND	1
(7)	ND	1
(8)	ND	1
(9)	ND	1
(10)	ND	1
(11)	ND	1
(12)	ND	1
(13)	ND	1
(14)	ND	1
(15)	ND	1
(16)	ND	1
(17)	ND	1
(18)	ND	1
(19)	ND	1
(20)	ND	1
(21)	ND	1
(22)	ND	1
(23)	ND	1
(24)	ND	1
(25)	ND	1
(26)	ND	1

Remark: The above limit was quoted according to Australian Competition and Consumer Act 2010 with Consumer Protection Notice No. 11, 2011 - Permanent ban on children's products for prohibition on sale of certain products containing specified phthalate.

Detection Limit = 0.01%(w/w)
ND = Not Detected

Tested Components: See component list in the last section of this report.

Date Sample Received : 16 Dec, 2022

Testing Period : 16 Dec, 2022 to 23 Dec, 2022



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Tests Conducted

11 Toxic Elements Analysis

As per Australian / New Zealand standard on safety of toys AS/NZS ISO 8124 part 3: 2021, acid extraction method was used and toxic elements content were determined by inductively coupled argon plasma spectrometry.

	<u>Result (mg/kg)</u>							<u>Limit (mg/kg)</u>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	
Sol. Barium (Ba)	<5	<5	<5	<5	<5	<5	<5	1000
Sol. Lead (Pb)	<5	<5	<5	<5	<5	<5	<5	90
Sol. Cadmium (Cd)	<5	<5	<5	<5	<5	<5	<5	75
Sol. Antimony (Sb)	<5	<5	<5	<5	<5	<5	<5	60
Sol. Selenium (Se)	<5	<5	<5	<5	<5	<5	<5	500
Sol. Chromium (Cr)	<5	<5	<5	<5	<5	<5	<5	60
Sol. Mercury (Hg)	<5	<5	<5	<5	<5	<5	<5	60
Sol. Arsenic (As)	<2.5	<2.5	<2.5	<2.5	<2.5	<2.5	<2.5	25

	<u>Result (mg/kg)</u>							<u>Limit (mg/kg)</u>
	(8)	(9)	(10)	(11)	(12)	(13)	(14)	
Sol. Barium (Ba)	<5	<5	<5	<5	<5	<5	<5	1000
Sol. Lead (Pb)	<5	<5	<5	<5	<5	<5	<5	90
Sol. Cadmium (Cd)	<5	<5	<5	<5	<5	<5	<5	75
Sol. Antimony (Sb)	<5	<5	<5	<5	<5	<5	<5	60
Sol. Selenium (Se)	<5	<5	<5	<5	<5	<5	<5	500
Sol. Chromium (Cr)	<5	<5	<5	<5	<5	<5	<5	60
Sol. Mercury (Hg)	<5	<5	<5	<5	<5	<5	<5	60
Sol. Arsenic (As)	<2.5	<2.5	<2.5	<2.5	<2.5	<2.5	<2.5	25

	<u>Result (mg/kg)</u>							<u>Limit (mg/kg)</u>
	(15)	(16)	(17)	(18)	(19)	(20)	(21)	
Sol. Barium (Ba)	<5	<5	<5	<5	<5	<5	<5	1000
Sol. Lead (Pb)	<5	<5	<5	<5	<5	<5	<5	90
Sol. Cadmium (Cd)	<5	<5	<5	<5	<5	<5	<5	75
Sol. Antimony (Sb)	<5	<5	<5	<5	<5	<5	<5	60
Sol. Selenium (Se)	<5	<5	<5	<5	<5	<5	<5	500
Sol. Chromium (Cr)	<5	<5	<5	<5	<5	<5	<5	60
Sol. Mercury (Hg)	<5	<5	<5	<5	<5	<5	<5	60
Sol. Arsenic (As)	<2.5	<2.5	<2.5	<2.5	<2.5	<2.5	<2.5	25

	<u>Result (mg/kg)</u>							<u>Limit (mg/kg)</u>
	(22)	(23)	(24)	(25)	(26)	(27)		
Sol. Barium (Ba)	<5	<5	<5	<5	<5	<5		1000
Sol. Lead (Pb)	<5	<5	<5	<5	<5	<5		90
Sol. Cadmium (Cd)	<5	<5	<5	<5	<5	<5		75
Sol. Antimony (Sb)	<5	<5	<5	<5	<5	<5		60
Sol. Selenium (Se)	<5	<5	<5	<5	<5	<5		500
Sol. Chromium (Cr)	<5	<5	<5	<5	<5	<5		60
Sol. Mercury (Hg)	<5	<5	<5	<5	<5	<5		60
Sol. Arsenic (As)	<2.5	<2.5	<2.5	<2.5	<2.5	<2.5		25

Remark: Sol. = Soluble
mg/kg = Milligram per kilogram



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Tests Conducted

Tested Components: See component list in the last section of this report.

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Testing Period : 16 Dec, 2022 to 23 Dec, 2022

12 Detection of Amines Derived from Azocolourants and Azodyes

By Gas Chromatographic - Mass Spectrometric (GC-MS) And High Performance Liquid Chromatographic (HPLC) Analysis.

Test Method: EN ISO 14362-1: 2017 for Textile Material

	Forbidden	Cas No.	Result (ppm)	
			Method T (27)	Method D (27)
1.	4-Aminodiphenyl	92-67-1	N	N
2.	Benzidine	92-87-5	N	N
3.	4-Chloro-o-Toluidine	95-69-2	N	N
4.	2-Naphthylamine	91-59-8	N	N
5.	o-Aminoazotoluene	97-56-3	N	N
6.	2-Amino-4-Nitrotoluene	99-55-8	N	N
7.	p-Chloroaniline	106-47-8	N	N
8.	2,4-Diaminoanisole	615-05-4	N	N
9.	4,4'-Diaminodiphenylmethane	101-77-9	N	N
10.	3,3'-Dichlorobenzidine	91-94-1	N	N
11.	3,3'-Dimethoxybenzidine	119-90-4	N	N
12.	3,3'-Dimethylbenzidine	119-93-7	N	N
13.	3,3'-Dimethyl-4,4'-diaminodiphenylmethane	838-88-0	N	N
14.	p-Cresidine	120-71-8	N	N
15.	4,4'-Methylene-Bis(2-Chloroaniline)	101-14-4	N	N
16.	4,4'-Oxydianiline	101-80-4	N	N
17.	4,4'-Thiodianiline	139-65-1	N	N
18.	o-Toluidine	95-53-4	N	N
19.	2,4-Toluylenediamine	95-80-7	N	N
20.	2,4,5-Trimethylaniline	137-17-7	N	N
21.	o-Anisidine	90-04-0	N	N
22.	p-Aminoazobenzene	60-09-3	N	N

Remark: N = Not Detected

Detection Limit = 5 ppm

Requirement = 30 ppm (Max.)

ppm = Parts per million = mg/kg

Method T: Direct buffer extraction as per EN ISO 14362-1: 2017 Section 10.2

Method D: Colourant extraction with Xylene as per EN ISO 14362-1: 2017 Section 10.1

Tested Components: See component list in the last section of this report.

Date Sample Received : 16 Dec, 2022

Testing Period : 16 Dec, 2022 to 23 Dec, 2022



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Tests Conducted

Photo



The Samples Were Submitted By The Client, Only For Reference.



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Tests Conducted



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Number: SHAH01530119

Tests Conducted

Components List:

- (1) Red plastic.
- (2) White plastic.
- (3) Green plastic.
- (4) Pink plastic.
- (5) Blue plastic.
- (6) Yellow plastic.
- (7) Black plastic.
- (8) Orange plastic.
- (9) Beige plastic.
- (10) Purple plastic.
- (11) Black plastic(front fence).
- (12) Transparent plastic(front light).
- (13) Red transparent plastic(tail light).
- (14) Black plastic(button on music player).
- (15) Red plastic(steering wheel button).
- (16) Grey plastic(door, door lock, exhaust funnel).
- (17) Black plastic(seat).
- (18) Black plastic(accelerator pedal).
- (19) Black plastic(wheel).
- (20) Black soft plastic(wire covering).
- (21) Red soft plastic(wire covering).
- (22) Yellow soft plastic(wire covering).
- (23) Black soft plastic(wheel antiskid part).
- (24) Bright silver grey coating on plastic(front fence, instrument panel).
- (25) Black coating on metal(chassis).
- (26) White adhesive plastic film with black printing(tail body).
- (27) Black webbing(safety belt).

End Of Report

The statements of conformity reported have considered the decision rule agreed, namely that Intertek have taken account of measurement uncertainty as calculated by Intertek, and applied according to ILAC-G8/09:2019 (Non-binary acceptance based on guard band $w = U$) except designation from the customer, regulation or test specification. This decision rule only applies to the numeric test results.

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